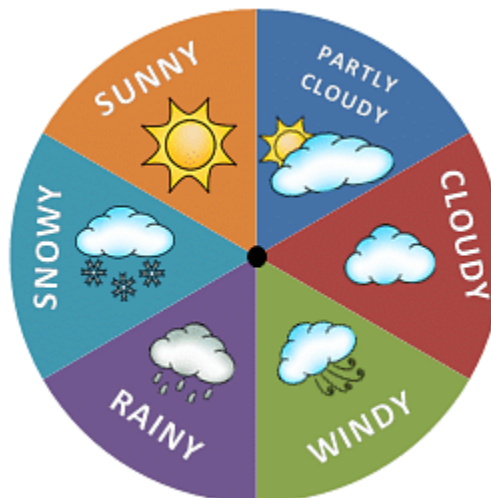


Weather and Climate

What is Weather?

The day-to-day conditions of the atmosphere at a place with respect to elements like humidity, temperature, wind speed, rainfall, etc. is called the weather of that place.

- Weather can be cloudy, sunny, rainy, stormy, or clear. It is a part of the natural phenomenon which maintains the equilibrium in the atmosphere.



What is the Weather like today?

- But conditions can be worse sometimes. When the atmospheric conditions are extreme or intense enough to cause property loss or life loss, such weather is termed severe weather.
- Changes in Weather Conditions give rise to seasons.
- In India, we enjoy three seasons: Summer, Winter & Monsoon

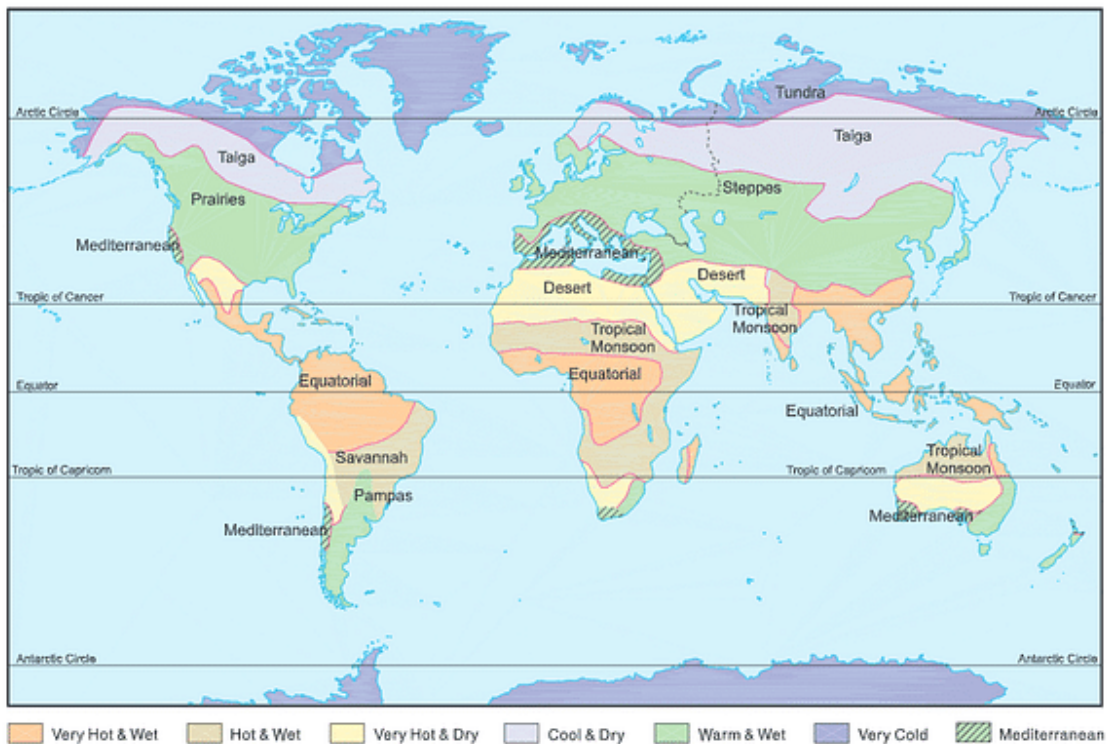


Seasons of India

What is Climate?

The weather conditions that prevail over a large area and for a long period of time is called climate.

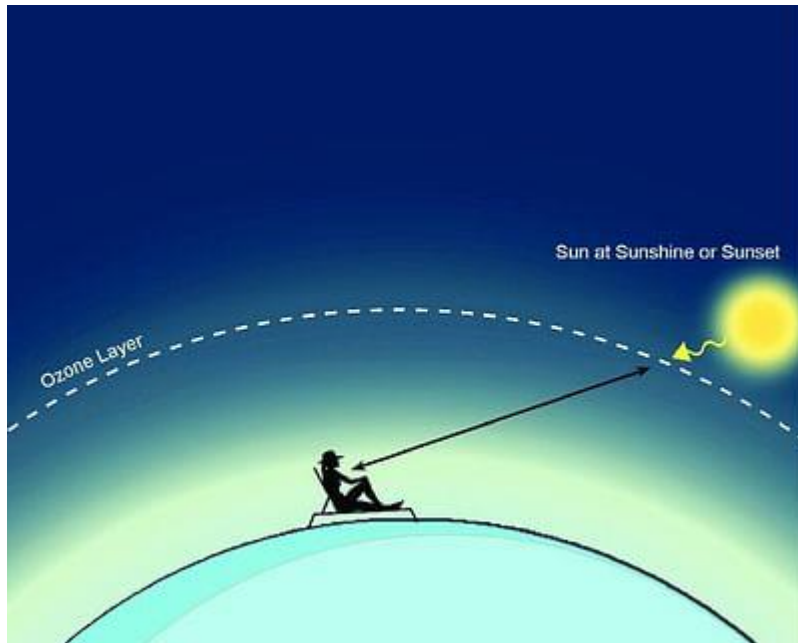
- The climate of a place remains more or less the same, year after year.
Different parts of the world have different types of climate.
- On the basis of the prevailing climatic conditions, the world can be divided into seven major climatic regions.



The seven major climatic regions of the world

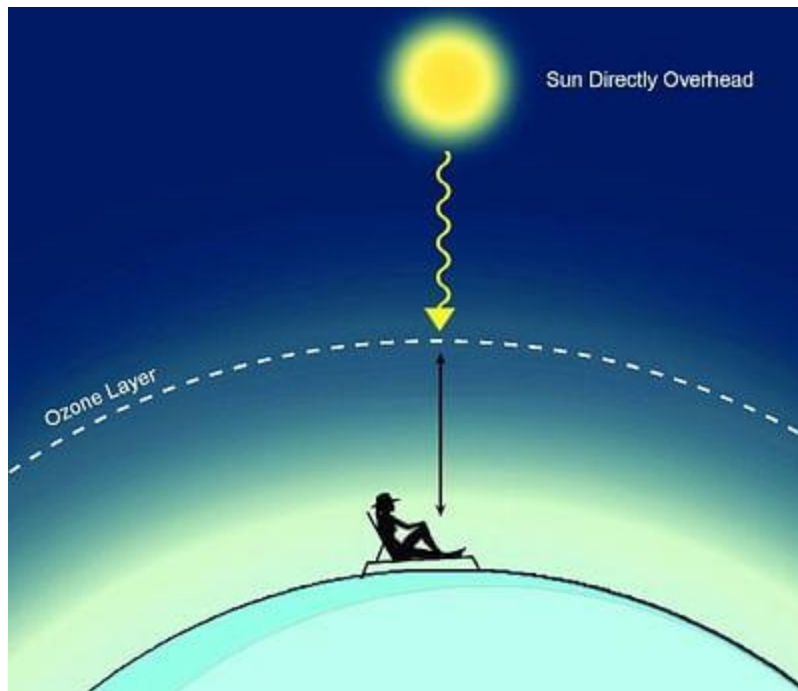
Why it is hot during the day and cool during the nights?

- It can be best understood once we know how we receive heat from the sun. It is cool in the morning.
- As the day progresses, the weather gets hotter. It is hottest by noontime. Then the day starts getting cool once again.



In morning and evening, sun rays are slanting and spread over a larger area making nights cool.

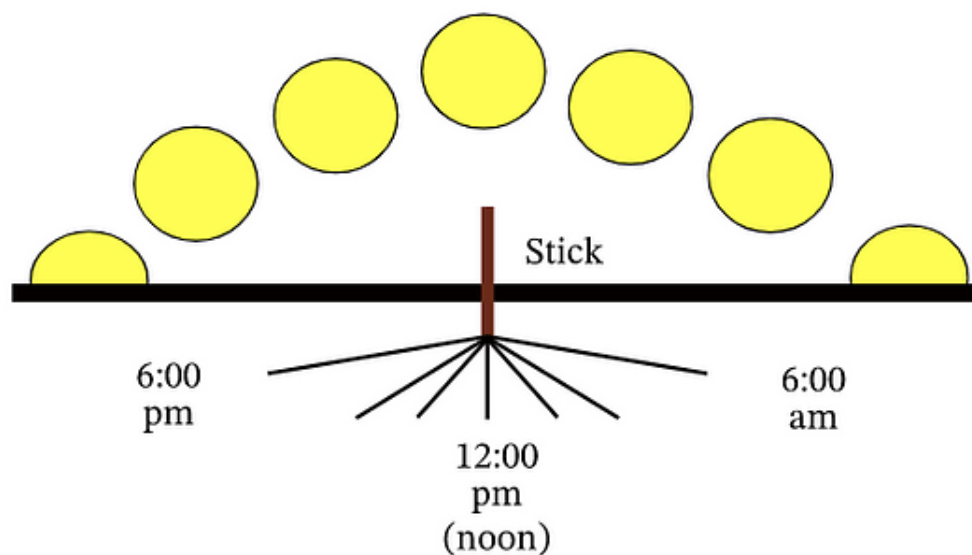
- Nights are cool. In the morning, the sun's rays are slanting and spread over a larger area. The same is the position in the evening. The spread of sun's rays over a large area leads to low temperatures.
- At noon, the sun is directly over our heads. The rays of the sun fall over a small area. The temperatures at noontime are therefore higher.



At noon, the sun is directly over our heads so, we feel hot during the day.

Why are shadows longer in the morning?

Shadows are longer during the morning and evening hours. That is so because the sun's rays are slanting. At noon the shadows are smallest.



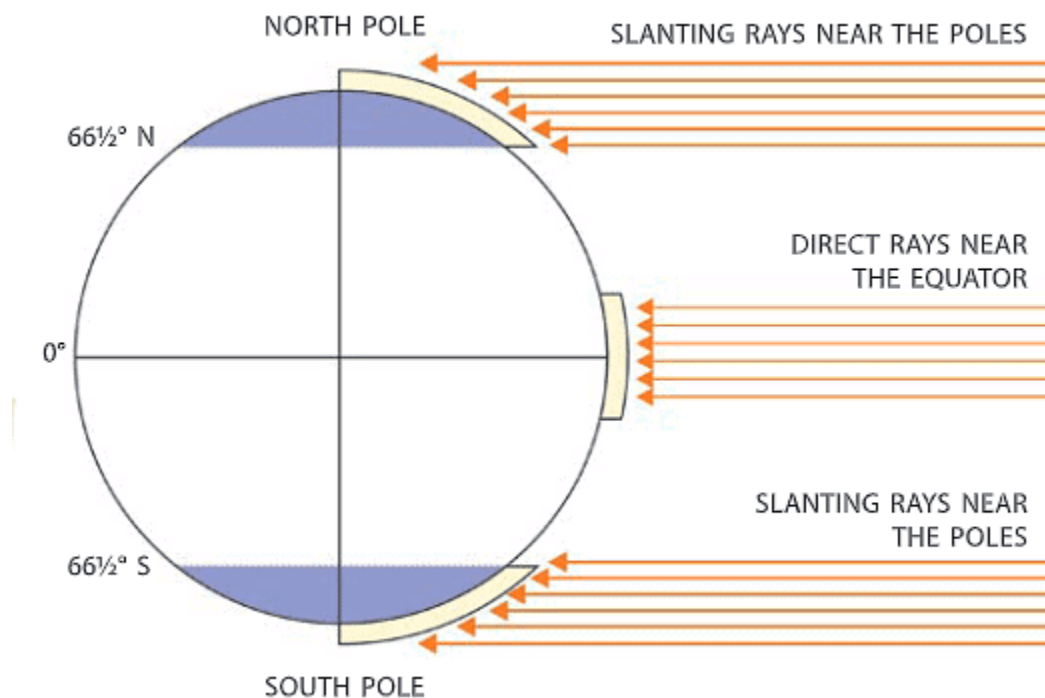
Shadows are longest during morning and evening hours.

What influences Climate?

Many factors influence the distribution of heat on Earth, thereby affecting the climate. Let us read about them.

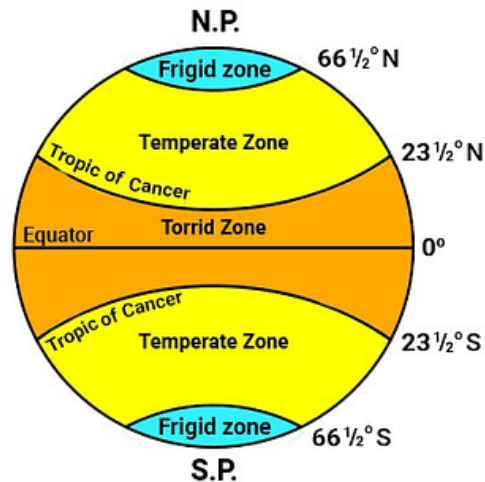
Distance from the Equator

- The sun's rays fall directly over the regions that are nearest to the equator. They cover a small area.
- As such, regions near the equator are hotter. The position is different at places near the poles.
- The rays of the sun are spread over a much larger area. Here they strike in a slanting fashion. So places near the poles are relatively colder.
- The areas between the poles and the Arctic and Antarctic circles are very cold.



How sun's rays fall on the earth.

- They are known as the Frigid Zones. Frigid means very cold. In contrast, the areas between the equator and the Tropics are very hot.
- Torrid means very hot and dry. From 23½° N to 23½° S the region gets maximum heat. It is known as the Torrid Zone. In between the Torrid and the Frigid zones, lies the area that is temperate zone.

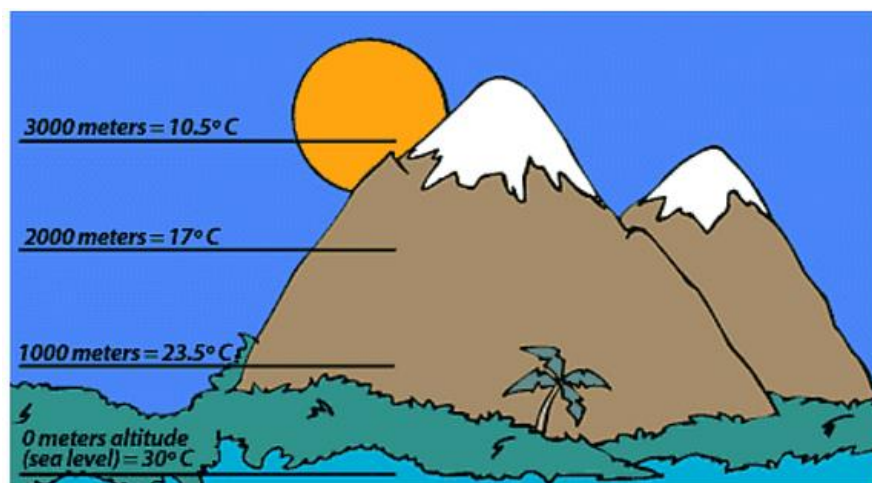


Temperature Zones of Earth

- This zone is neither very hot nor very cold. So we now have Rule I. Whether a region has a hot, temperate, or cold climate depends on its distance from the equator.

Altitude of Place

- Bengaluru is cooler than Chennai. The two places are nearly equidistant from the equator. The difference lies in their height above sea level – Altitude. Chennai is situated at sea level, while Bengaluru is about 550 metres above sea level.
- The air above the surface of earth works like layers of blankets. The more the layers, the hotter it would be at a particular place.
- If a place is situated at a higher altitude, there are fewer layers and hence the place is cooler. It is very cold at the high mountain peaks.

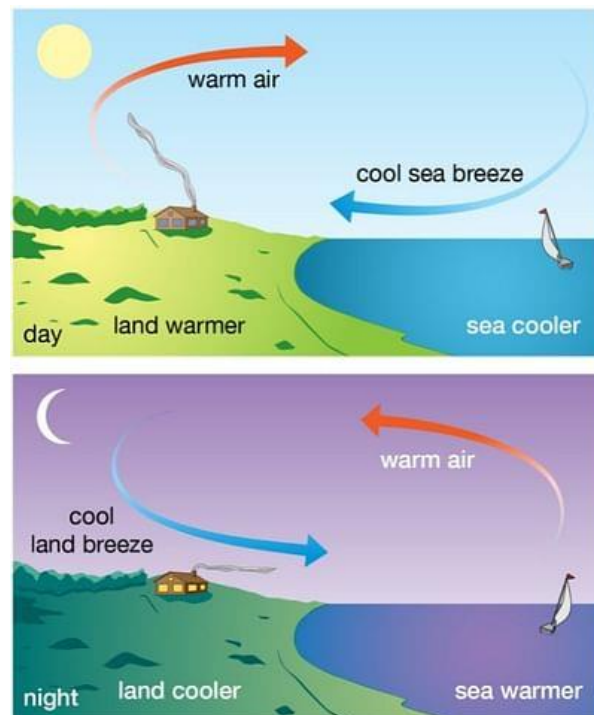


As height increases temperature decreases.

- The rivers of ice (glaciers) slowly move down the mountain slopes. As ice comes down to lower heights, where temperatures are higher, it melts forming rivers.
- River Ganga originates from the glacier Gangotri. When it is summertime in the plains, people move to hill stations to escape the heat.
- Ferozpur (Punjab) and Shimla (Himachal Pradesh) may have nearly the same latitude. Shimla (2000–2430 m above sea level) rarely has temperatures exceeding 25°C.
- Ferozpur, on the other hand, is very hot during the summer season. For every 165 meters above sea level, the temperature drops by 1°C. Now we have Rule II. The climate varies with altitude or the height above sea level.

Distance from Sea

- Patna in Bihar is farther from the Equator as compared to Kolkata in West Bengal. And yet Patna is hotter than Kolkata. The difference arises because Kolkata is situated near the sea. During day time, the land gets heated more quickly than water. The air over land surface becomes lighter and rises up creating a low-pressure zone.
- Breezes from the neighboring sea blow to fill up the low-pressure zone on the coast. The moist sea breezes moderate the heat over the land surface.



The sea has a moderating influence on the region close to it.

- The reverse happens at night. The movement of breezes from the sea during the day and from land to sea during nights makes the climate MODERATE on the seashores. Places far off from the sea have extreme climates.
- For that reason, Patna is very hot during summers and cold during winters. The seaports are neither very hot during summers nor very cold during the winters. Rule III tells us that **Distance from Sea makes climate moderate or extreme.**

The direction of the winds

- If the place or region gets winds from a hot place, it will become hot.
- If it receives winds from a cold place it will become cold.

Humidity

- Humidity is related to the amount of moisture in the air. It depends on the distance of the place from any large water body such as a lake, or a river or a sea. It also depends on the wind direction.
- A place close to any large water body will have more moisture in the air hence the climate will be humid.



Places that are humid receives a lot of rainfall.

- Winds carrying more moisture for any reason will make the place humid. Places getting rains in the Monsoon season are humid in that season.

Rainfall

- How much a place receives rainfall also affects its climate. Some places get rainfall almost all around the year and others may not get any rainfall at all. In both cases, climates are going to be different.



- The climate of a place is determined by some of the reasons working together or one of the reasons which make other reasons to come together to decide its climate.
- It is always interesting to look for the reasons which have determined the climate of a particular place or region.

Keep in Mind!

- Weather is the condition of the atmosphere of a place at a given time.
- Climate is the average of the weather conditions over a year.
- Factors that influence Weather and climate are:
 - (i) Distance from the Equator
 - (ii) Altitude
 - (iii) Distance from the Sea
 - (iv) Direction of the Winds
 - (v) Humidity and
 - (vi) Rainfall